

2_DataViewing_Basics

February 11, 2025

```
[64]: pip install pandas
```

```
Requirement already satisfied: pandas in ./lib/python3.13/site-packages (2.2.3)
Requirement already satisfied: numpy>=1.26.0 in ./lib/python3.13/site-packages
(from pandas) (2.2.1)
Requirement already satisfied: python-dateutil>=2.8.2 in ./lib/python3.13/site-
packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in ./lib/python3.13/site-packages
(from pandas) (2024.2)
Requirement already satisfied: tzdata>=2022.7 in ./lib/python3.13/site-packages
(from pandas) (2024.2)
Requirement already satisfied: six>=1.5 in ./lib/python3.13/site-packages (from
python-dateutil>=2.8.2->pandas) (1.17.0)
Note: you may need to restart the kernel to use updated packages.
```

```
[65]: import pandas as pd
```

```
[66]: df = pd.read_csv('movies.csv')
```

```
[67]: df.head()
```

```
[67]:
```

		title	industry	release_year	\
0		Pather Panchali	Bollywood	1955	
1		Doctor Strange in the Multiverse of Madness	Hollywood	2022	
2		Thor: The Dark World	Hollywood	2013	
3		Thor: Ragnarok	Hollywood	2017	
4		Thor: Love and Thunder	Hollywood	2022	

	imdb_rating		studio	budget	revenue	unit	\
0	8.3	Government of West Bengal		70000.0	100000.0	Thousands	
1	7.0		Marvel Studios	200.0	954.8	Millions	
2	6.8		Marvel Studios	165.0	644.8	Millions	
3	7.9		Marvel Studios	180.0	854.0	Millions	
4	6.8		Marvel Studios	250.0	670.0	Millions	

	currency	language
0	INR	Bengali
1	USD	English

```

2      USD  English
3      USD  English
4      USD  English

```

```
[68]: df.shape
```

```
[68]: (37, 10)
```

```
[69]: df.columns
```

```
[69]: Index(['title', 'industry', 'release_year', 'imdb_rating', 'studio', 'budget',
          'revenue', 'unit', 'currency', 'language'],
          dtype='object')
```

```
[70]: df.industry.unique()
```

```
[70]: array(['Bollywood', 'Hollywood'], dtype=object)
```

```
[71]: df.language.value_counts()
```

```
[71]: language
English      20
Hindi        12
Telugu        3
Bengali        1
Kannada        1
Name: count, dtype: int64
```

```
[72]: df.industry.value_counts()
```

```
[72]: industry
Hollywood      20
Bollywood      17
Name: count, dtype: int64
```

```
[73]: df[['title', 'imdb_rating']]
```

```
[73]:
```

	title	imdb_rating
0	Pather Panchali	8.3
1	Doctor Strange in the Multiverse of Madness	7.0
2	Thor: The Dark World	6.8
3	Thor: Ragnarok	7.9
4	Thor: Love and Thunder	6.8
5	The Shawshank Redemption	9.3
6	Interstellar	8.6
7	The Pursuit of Happyness	8.0
8	Gladiator	8.5
9	Titanic	7.9

10	It's a Wonderful Life	8.6
11	Avatar	7.8
12	The Godfather	9.2
13	The Dark Knight	9.0
14	Schindler's List	9.0
15	Jurassic Park	8.2
16	Parasite	8.5
17	Avengers: Endgame	8.4
18	Avengers: Infinity War	8.4
19	Captain America: The First Avenger	6.9
20	Captain America: The Winter Soldier	7.8
21	Dilwale Dulhania Le Jayenge	8.0
22	3 Idiots	8.4
23	Kabhi Khushi Kabhie Gham	7.4
24	Bajirao Mastani	7.2
25	Taare Zameen Par	8.3
26	Munna Bhai M.B.B.S.	8.1
27	PK	8.1
28	Sanju	NaN
29	The Kashmir Files	8.3
30	Bajrangi Bhaijaan	8.1
31	Race 3	1.9
32	Shershaah	8.4
33	K.G.F: Chapter 2	8.4
34	Pushpa: The Rise - Part 1	7.6
35	RRR	8.0
36	Baahubali: The Beginning	8.0

```
[74]: df[(df.release_year >= 2000) & (df.release_year<=2010)]
```

[74]:	title	industry	release_year	imdb_rating	\
7	The Pursuit of Happyness	Hollywood	2006	8.0	
8	Gladiator	Hollywood	2000	8.5	
11	Avatar	Hollywood	2009	7.8	
13	The Dark Knight	Hollywood	2008	9.0	
22	3 Idiots	Bollywood	2009	8.4	
23	Kabhi Khushi Kabhie Gham	Bollywood	2001	7.4	
25	Taare Zameen Par	Bollywood	2007	8.3	
26	Munna Bhai M.B.B.S.	Bollywood	2003	8.1	

	studio	budget	revenue	unit	currency	language
7	Columbia Pictures	55.0	307.1	Millions	USD	English
8	Universal Pictures	103.0	460.5	Millions	USD	English
11	20th Century Fox	237.0	2847.0	Millions	USD	English
13	Syncopy	185.0	1006.0	Millions	USD	English
22	Vinod Chopra Films	550.0	4000.0	Millions	INR	Hindi
23	Dharma Productions	390.0	1360.0	Millions	INR	Hindi

25		NaN	120.0	1350.0	Millions	INR	Hindi
26	Vinod Chopra Productions		100.0	410.0	Millions	INR	Hindi

```
[75]: df.loc[df['studio'] == 'Marvel Studios',['title','industry','studio']]
```

```
[75]:
```

	title	industry	studio
1	Doctor Strange in the Multiverse of Madness	Hollywood	Marvel Studios
2	Thor: The Dark World	Hollywood	Marvel Studios
3	Thor: Ragnarok	Hollywood	Marvel Studios
4	Thor: Love and Thunder	Hollywood	Marvel Studios
17	Avengers: Endgame	Hollywood	Marvel Studios
18	Avengers: Infinity War	Hollywood	Marvel Studios
19	Captain America: The First Avenger	Hollywood	Marvel Studios
20	Captain America: The Winter Soldier	Hollywood	Marvel Studios

```
[76]: df.describe()
```

```
[76]:
```

	release_year	imdb_rating	budget	revenue
count	37.000000	36.000000	37.000000	37.000000
mean	2007.027027	7.919444	2084.975135	4117.135135
std	17.657995	1.204947	11477.487145	16372.462682
min	1946.000000	1.900000	1.000000	3.100000
25%	2001.000000	7.800000	15.500000	263.100000
50%	2014.000000	8.100000	165.000000	701.800000
75%	2018.000000	8.400000	250.000000	2000.000000
max	2022.000000	9.300000	70000.000000	100000.000000

```
[77]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 37 entries, 0 to 36
Data columns (total 10 columns):
#   Column          Non-Null Count  Dtype
---  -
0   title           37 non-null    object
1   industry        37 non-null    object
2   release_year    37 non-null    int64
3   imdb_rating     36 non-null    float64
4   studio          34 non-null    object
5   budget          37 non-null    float64
6   revenue         37 non-null    float64
7   unit            37 non-null    object
8   currency        37 non-null    object
9   language        37 non-null    object
dtypes: float64(3), int64(1), object(6)
memory usage: 3.0+ KB
```

```
[78]: df.imdb_rating.max()
```

```
[78]: np.float64(9.3)
```

```
[79]: df[(df.imdb_rating == df.imdb_rating.max() )| (df.imdb_rating == df.imdb_rating.
↪min())]
```

```
[79]:
```

	title	industry	release_year	imdb_rating	\
5	The Shawshank Redemption	Hollywood	1994	9.3	
31	Race 3	Bollywood	2018	1.9	

	studio	budget	revenue	unit	currency	language
5	Castle Rock Entertainment	25.0	73.3	Millions	USD	English
31	Salman Khan Films	1.8	3.1	Billions	INR	Hindi

```
[80]: df['Age'] = 2025 - df.release_year
```

```
[81]: df.head()
```

```
[81]:
```

	title	industry	release_year	\
0	Pather Panchali	Bollywood	1955	
1	Doctor Strange in the Multiverse of Madness	Hollywood	2022	
2	Thor: The Dark World	Hollywood	2013	
3	Thor: Ragnarok	Hollywood	2017	
4	Thor: Love and Thunder	Hollywood	2022	

	imdb_rating	studio	budget	revenue	unit	\
0	8.3	Government of West Bengal	70000.0	100000.0	Thousands	
1	7.0	Marvel Studios	200.0	954.8	Millions	
2	6.8	Marvel Studios	165.0	644.8	Millions	
3	7.9	Marvel Studios	180.0	854.0	Millions	
4	6.8	Marvel Studios	250.0	670.0	Millions	

	currency	language	Age
0	INR	Bengali	70
1	USD	English	3
2	USD	English	12
3	USD	English	8
4	USD	English	3

```
[82]: df['Profit'] = df.apply(lambda x: x['revenue'] - x['budget'], axis = 1)
df.head()
```

```
[82]:
```

	title	industry	release_year	\
0	Pather Panchali	Bollywood	1955	
1	Doctor Strange in the Multiverse of Madness	Hollywood	2022	
2	Thor: The Dark World	Hollywood	2013	
3	Thor: Ragnarok	Hollywood	2017	
4	Thor: Love and Thunder	Hollywood	2022	

	imdb_rating	studio	budget	revenue	unit \
0	8.3	Government of West Bengal	70000.0	100000.0	Thousands
1	7.0	Marvel Studios	200.0	954.8	Millions
2	6.8	Marvel Studios	165.0	644.8	Millions
3	7.9	Marvel Studios	180.0	854.0	Millions
4	6.8	Marvel Studios	250.0	670.0	Millions

	currency	language	Age	Profit
0	INR	Bengali	70	30000.0
1	USD	English	3	754.8
2	USD	English	12	479.8
3	USD	English	8	674.0
4	USD	English	3	420.0

```
[83]: df.index
```

```
[83]: RangeIndex(start=0, stop=37, step=1)
```

```
[84]: df.set_index("title", inplace = True) #inplace updates the actual dataframe
```

```
[85]: df.index
```

```
[85]: Index(['Pather Panchali', 'Doctor Strange in the Multiverse of Madness',
          'Thor: The Dark World ', 'Thor: Ragnarok ', 'Thor: Love and Thunder ',
          'The Shawshank Redemption', 'Interstellar', 'The Pursuit of Happyness',
          'Gladiator', 'Titanic', 'It's a Wonderful Life', 'Avatar',
          'The Godfather', 'The Dark Knight', 'Schindler's List', 'Jurassic Park',
          'Parasite', 'Avengers: Endgame', 'Avengers: Infinity War',
          'Captain America: The First Avenger',
          'Captain America: The Winter Soldier', 'Dilwale Dulhania Le Jayenge',
          '3 Idiots', 'Kabhi Khushi Kabhie Gham', 'Bajirao Mastani ',
          'Taare Zameen Par', 'Munna Bhai M.B.B.S.', 'PK', 'Sanju',
          'The Kashmir Files', 'Bajrangi Bhaijaan', 'Race 3', 'Shershaah',
          'K.G.F: Chapter 2', 'Pushpa: The Rise - Part 1', 'RRR',
          'Baahubali: The Beginning'],
          dtype='object', name='title')
```

```
[86]: df.head()
```

```
[86]:
```

title	industry	release_year \
Pather Panchali	Bollywood	1955
Doctor Strange in the Multiverse of Madness	Hollywood	2022
Thor: The Dark World	Hollywood	2013
Thor: Ragnarok	Hollywood	2017
Thor: Love and Thunder	Hollywood	2022

	imdb_rating \
title	
Pather Panchali	8.3
Doctor Strange in the Multiverse of Madness	7.0
Thor: The Dark World	6.8
Thor: Ragnarok	7.9
Thor: Love and Thunder	6.8

	studio \
title	
Pather Panchali	Government of West Bengal
Doctor Strange in the Multiverse of Madness	Marvel Studios
Thor: The Dark World	Marvel Studios
Thor: Ragnarok	Marvel Studios
Thor: Love and Thunder	Marvel Studios

	budget	revenue	unit \
title			
Pather Panchali	70000.0	100000.0	Thousands
Doctor Strange in the Multiverse of Madness	200.0	954.8	Millions
Thor: The Dark World	165.0	644.8	Millions
Thor: Ragnarok	180.0	854.0	Millions
Thor: Love and Thunder	250.0	670.0	Millions

	currency	language	Age	Profit
title				
Pather Panchali	INR	Bengali	70	30000.0
Doctor Strange in the Multiverse of Madness	USD	English	3	754.8
Thor: The Dark World	USD	English	12	479.8
Thor: Ragnarok	USD	English	8	674.0
Thor: Love and Thunder	USD	English	3	420.0

```
[93]: df.loc[['Pather Panchali','Doctor Strange in the Multiverse of Madness']]
```

```
[93]:
```

	industry	release_year \
title		
Pather Panchali	Bollywood	1955
Doctor Strange in the Multiverse of Madness	Hollywood	2022

	imdb_rating \
title	
Pather Panchali	8.3
Doctor Strange in the Multiverse of Madness	7.0

	studio \
title	
Pather Panchali	Government of West Bengal

Doctor Strange in the Multiverse of Madness	Marvel Studios			
	budget	revenue	unit	\
title				
Pather Panchali	70000.0	100000.0	Thousands	
Doctor Strange in the Multiverse of Madness	200.0	954.8	Millions	
	currency	language	Age	Profit
title				
Pather Panchali	INR	Bengali	70	30000.0
Doctor Strange in the Multiverse of Madness	USD	English	3	754.8

```
[95]: df.iloc[2:5]
```

```
[95]:
```

	industry	release_year	imdb_rating	studio	\		
title							
Thor: The Dark World	Hollywood	2013	6.8	Marvel Studios			
Thor: Ragnarok	Hollywood	2017	7.9	Marvel Studios			
Thor: Love and Thunder	Hollywood	2022	6.8	Marvel Studios			
	budget	revenue	unit	currency	language	Age	\
title							
Thor: The Dark World	165.0	644.8	Millions	USD	English	12	
Thor: Ragnarok	180.0	854.0	Millions	USD	English	8	
Thor: Love and Thunder	250.0	670.0	Millions	USD	English	3	
	Profit						
title							
Thor: The Dark World	479.8						
Thor: Ragnarok	674.0						
Thor: Love and Thunder	420.0						

```
[96]: df.reset_index(inplace = True)
```

```
[ ]:
```